Hon Tik Tse

Ph.D. Student | University of Alberta
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Education

- **Ph.D. in Computing Science** — Sep 2024 - Present
  University of Alberta — Edmonton, AB, Canada
  - Supervisor: Marlos C. Machado
- **Master of Science in Computer Science** — Aug 2022 - May 2024
  University of Massachusetts Amherst — Amherst, MA, USA
- **Bachelor of Science in Computer Science with Honours, First Class** — Aug 2018 - Jul 2022
  The Chinese University of Hong Kong — Hong Kong

Research Experience

- **University of Alberta**
  Ph.D. Research advised by Marlos C. Machado — Sep 2024 - Present
  - Laid the foundation for applying the default representation, a temporal representation capturing both transition and reward dynamics, in reinforcement learning [Corresponding paper accepted at NeurIPS 2025]
  - Proposed an objective for directly learning the principal eigenvector of the default representation using a neural network [Corresponding paper in submission]
- **University of Massachusetts Amherst**
  Research Project advised by Philip Thomas & Scott Niekum — June 2023 - May 2024
  - Proposed an algorithm for high confidence policy improvement in the reinforcement learning from human feedback (RLHF) setting, where a reward function is not available and only preferences over demonstrations are available [Corresponding paper accepted at RLC 2025]
- **University of Massachusetts Amherst**
  Research Project advised by Hava Siegelmann — Feb 2023 - May 2023
  - Adapted Deep Q-network (DQN) for a biologically plausible forward learning framework called Signal Propagation, achieving comparable performance as standard DQN in a few simple domains
- **The Chinese University of Hong Kong**
  Senior Year Research advised by Ho-fung Leung — Sep 2021 - May 2022
  - Extended ϵ-greedy to a hierarchical action space in multi-agent reinforcement learning [Corresponding paper available on arXiv]

Publications

- Peer-Reviewed
  - **DROGO: Default Representation Objective via Graph Optimization in Reinforcement Learning**
    Hon Tik Tse, Marlos C. Machado
    Advances in Neural Information Processing Systems **(NeurIPS)**, 2026.
  - **Reward-Aware Proto-Representations in Reinforcement Learning**
    Hon Tik Tse, Siddarth Chandrasekar, Marlos C. Machado
    Advances in Neural Information Processing Systems **(NeurIPS)**, 2025.
  - **High-Confidence Policy Improvement from Human Feedback**
    Hon Tik Tse, Philip S. Thomas, Scott Niekum
    Reinforcement Learning Journal **(RLJ)**, 2025.
- In Submission
  - **The Reward-Aware Laplacian Representation in Zero-Shot Reinforcement Learning**
    Hon Tik Tse, Siddarth Chandrasekar, Dikshant Shehmar, Marlos C. Machado
- Preprint
  - **Exploiting Semantic Epsilon Greedy Exploration Strategy in Multi-Agent Reinforcement Learning**
    Hon Tik Tse, Ho-fung Leung
    CoRR, abs/2201.10803, 2022.

Honors and Awards

- University of Alberta
  - PhD Early Achievement Award 2026
  - Graduate Recruitment Scholarship 2025
- The Chinese University of Hong Kong
  - Shaw College Young Scholars 2022
  - Professor S.C. Loh Scholarship for Computer Science and Engineering 2021, 2022
  - Faculty of Engineering Dean's List 2019-2022
  - Professor Charles K. Kao Research Exchange Scholarship 2021
  - Outstanding Academic Performance - Gold Award 2021

Teaching

- Course Development
  - CMPUT 628: Deep Reinforcement Learning Winter 2026
- Teaching Assistant
  - CMPUT 365: Introduction to Reinforcement Learning (Head TA) Fall 2026
  - CMPUT 291: Introduction to File and Database Management Winter 2025

Talks

- Reward-Aware Proto-Representations in Reinforcement Learning Virtual Talk, NeurIPS 2025
- High-Confidence Policy Improvement from Human Feedback RLC, Aug 7 2025
- Reward-Aware Proto-Representations in Reinforcement Learning Amii AI Seminar, 25 July 2025
- Reward-Aware Proto-Representations in Reinforcement Learning RLAI Tea Time Talk, 23 July 2025

Service

- Reviewer: ICML (2026 [Silver Reviewer Award]), NeurIPS (2026), ICLR (2026,2027), RLC (2025,2026), TMLR (2025,2026)

Skills

- Programming Languages: Python (PyTorch, Jax)